



THE KENYA NATIONAL EXAMINATIONS COUNCIL

SCHOOL BASED ASSESSMENT

INTEGRATED SCIENCE

Grade 7 – 2026

LEARNER'S NAME: _____

ASSESSMENT NUMBER: _____ DATE: _____

SCHOOL NAME: _____

INSTRUCTIONS TO LEARNERS

1. *This paper consists of 7 questions.*
2. *Answer **ALL** the questions in the spaces provided.*

The grid below to be filled by the teacher after marking the learners' work. During uploading, the teacher to upload the scores per task.

Tasks	Scientific Investigation	Matter, Elements and Compounds	Living Things	Force and Energy
Question Numbers	1	2-3	4-5	6 -7
Max. Score	9	8	7	6
Learner's Score				

This paper has 6 printed pages

TURN OVER

1. Figure 1 shows a set up used by learners to investigate how the amount of sunlight affects the growth of bean plants.

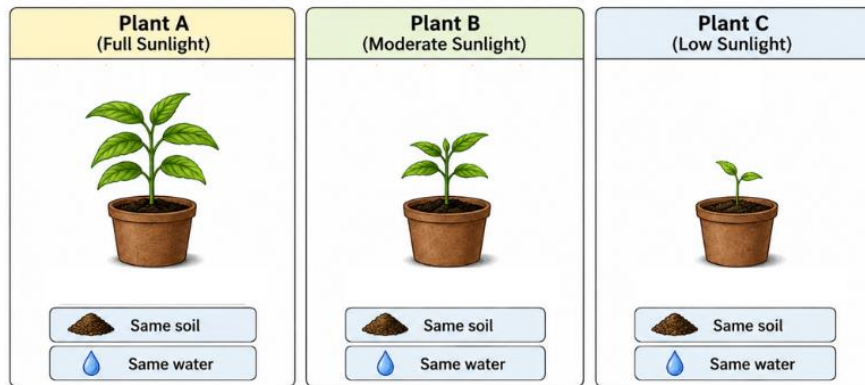


Figure 1

The learners planted three identical seedlings of bean plants at the same time and placed them in different locations.

a) (i) State the factor affecting growth of plants that was changed in this investigation?

(1 mark)

(ii) Explain your answer in (i) above.

(1 mark)

b) (i) Identify **two** factors that the learners did not change.

(2 marks)

(ii) Explain why keeping the factors you have identified in (i) constant was important.

(1 mark)

c) After four weeks, plant **A** grew 18 cm, Plant **B** grew 12 cm, and Plant **C** grew 5 cm.

(i) State the conclusion the learners made from these results.

(1 mark)

(ii) Give a reason for your answer in (i).

(1 mark)

- d) A learner suggests watering plant **C** more than the other plants to help it grow faster. Explain how it would affect the aim of the investigation. (2 marks)

2. Figure 2 shows layers of sand, salt, iron filings, and small plastic beads in one container.

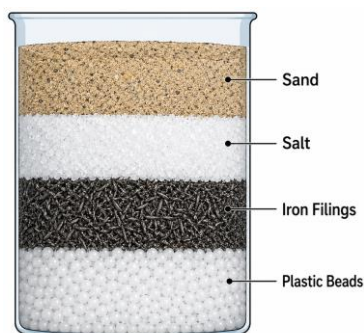


Figure 2

A learner needs to recover each substance separately for reuse.

- a) The learner decides to start the separation process by adding water to the mixture.

Give a reason whether the learner's decision is the best first step. (1 mark)

- b) Suppose plastic beads float on water while sand sinks. Describe the steps you would use to separate and recover all the four substances. (2 marks)

3. A learner is investigating three colourless solutions **P**, **Q** and **R**. The learner does not know which solution is acidic, basic or neutral. The following observations are recorded:

Solution	Blue litmus paper	Red litmus paper
P	Turns red	Remains red
Q	Remains blue	Turns blue
R	Remains blue	Remains red

The learner makes the following claim:

“Solution **R** must be a base because neither litmus paper changes colour.”

a) Explain whether the learner was right. (1 mark)

b) The learner accidentally mixes equal amounts of **P** and **Q**.

(i) Predict how the acidity of the resulting mixture will change. (1 mark)

(ii) Explain your answer in (i) using the properties of **P** and **Q**. (1 mark)

c) A learner suggests tasting **P**, **Q** and **R** to identify them. Why is this method unsafe?

Suggest a safer scientific method for identifying the solutions. (1 mark)

d) The teacher says, "Litmus paper alone may not always provide enough information about how strongly acidic or basic a solution is." Explain why this statement is correct. (1 mark)

4. During a school health talk, two learners discuss the changes that occur during adolescence. A learner claims that facial hair growth in boys and the start of menstruation in girls are signs that the reproductive system has begun functioning.

a) Give one piece of evidence from the scenario that supports this claim. (1 mark)

b) Explain why menstruation occurs only in girls and not in boys. (1 mark)

c) Describe the role of **one** reproductive organ whose function becomes active during adolescence. (1 mark)

5. During a hot sports day, two learners, **A** and **B**, take part in the same physical activity. Learner **A** drinks adequate water, while Learner **B** drinks very little water. At the end of the activity, Learner **B** passes a small amount of dark-coloured urine and feels thirsty.

a) State the role played by the kidney in maintaining the body's water balance. (1 mark)

b) Learner **B** produced a small amount of dark-coloured urine after drinking very little water. Explain how the kidneys' response to dehydration led to this observation. (1 mark)

c) Learner **B** says, “*The dark colour means my kidneys have produced more waste than usual.*”

(i) Do you agree? _____

(ii) Give a reason for your answer in (i). (1 mark)

d) Learner **B** continues losing water through sweating without replacing it. State one other way the body would respond to reduce further water loss. (1 mark)

6. Study the circuit diagrams below and use them to answer the questions.

Figure 2 — Circuit A

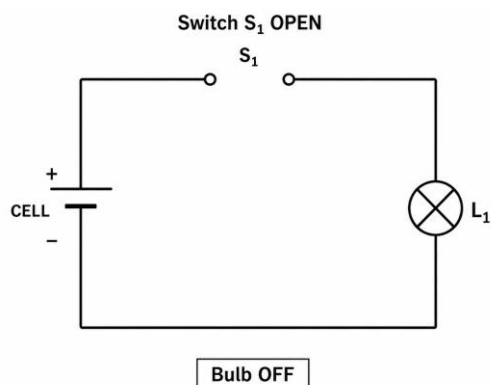
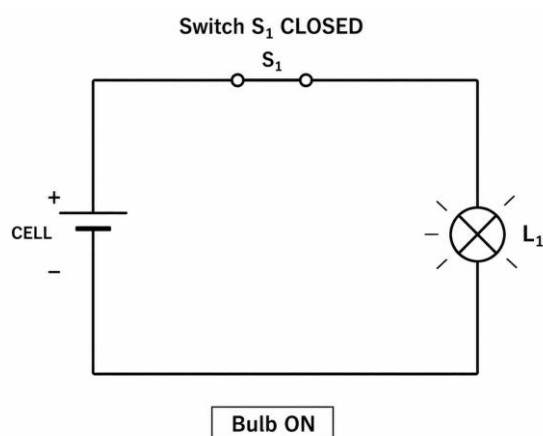


Figure 3 — Circuit B



A learner concludes:

“The switch produces electricity when it is closed.”

a) Explain whether the learner's conclusion is correct. (1 mark)

b) The learner then cuts the wire in Circuit **B**. Explain what will happen to the bulb even though the cell is still connected. (1 mark)

c) The learner wants to repair the cut wire in (b) above while the cell is still connected. Another learner says, "*It is safe.*"

Give a reason whether you agree with the statement. (1 mark)

7. Figure 4 shows two bar magnets **P** and **Q** placed by a learner on smooth surfaces. The magnets move as indicated.



Figure 4

a) Explain how the magnets moved apart even though they were not touching each other. (1 mark)

b) Explain how you would arrange magnets **P** and **Q** so that they move towards each other. (1 mark)

c) A learner is given a third magnet and asked to make magnets **P** and **Q** repel each other more strongly. Suggest **one** practical way of doing this and explain your answer. (1 mark)

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